

Challenge (Habanero)

- Q1.** A recipe for making bread requires a ratio of 2 : 3 of flour to water. If 6 cups of flour are used, how many cups of water are needed?
(A) 4
(B) 6
(C) 8
(D) 9
(E) 12
- Q2.** In a construction project, the ratio of concrete to sand is 3 : 2. If 12 tons of concrete are used, how many tons of sand are needed?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q3.** A baker is making a cake that requires a ratio of 1 : 2 of sugar to flour. If 3 cups of sugar are used, how many cups of flour are needed?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q4.** In a recipe for making cookies, the ratio of butter to sugar is 2 : 5. If 10 tablespoons of butter are used, how many tablespoons of sugar are needed?
(A) 10
(B) 12
(C) 20
(D) 25
(E) 30
- Q5.** A group of friends want to share some candy in a ratio of 3 : 4 : 5. If they have 120 pieces of candy, how many pieces will the person with the ratio 5 get?
(A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q6.** A builder is constructing a building with a height to width ratio of 5 : 3. If the height of the building is 50 meters, what is the width?
(A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q7.** A chef is making a sauce that requires a ratio of 2 : 3 of oil to vinegar. If 9 cups of oil are used, how many cups of vinegar are needed?
(A) 6
(B) 10
(C) 12
(D) 13.5
(E) 15
- Q8.** In a recipe for making a smoothie, the ratio of yogurt to fruit is 3 : 2. If 9 cups of yogurt are used, how many cups of fruit are needed?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10

Open Ended Questions (FRQ)

- Q9.** A construction company is building a bridge with a length to width ratio of 10 : 3. If the length of the bridge is 150 meters, what is the width? Show your work and explain your reasoning.
- Q10.** A recipe for making a cake requires a ratio of 2 : 3 : 5 of flour, sugar, and eggs. If 8 cups of flour are used, how many cups of sugar and eggs are needed? Show your work and explain your reasoning.

Chef's Tips

- Read each question carefully
- Use the given ratios to set up proportions
- Check your work and explain your reasoning